

Maharashtra Public Service Commission Assessment

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Asst Prof Group A Lecturer Group B Physics

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Asst Prof Group A Lecturer Group B Physics

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Consider the following operations in vector calculus.

- a. Gradient of scalar function
- b. Divergence of vector field
- c. Curl of vector field

The type of above quantities is respectively

Options :

- 1. Scalar, vector, scalar
- 2. Vector, vector, scalar
- 3. Vector, scalar, vector
- 4. Scalar, scalar, vector

The Fourier transform of the function $f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$ is given by

Options :

1. $\sqrt{\frac{2}{\pi}} \frac{\cos(sa)}{s}$

2. $\sqrt{\frac{2}{\pi}} \frac{\sin(sa)}{s}$

3. $\frac{2 \sin(sa)}{s}$

4. $\sqrt{\frac{\pi}{2}} \frac{\sin(sa)}{s}$

What will be the value of

$$\lim_{z \rightarrow i} \frac{z^{10} + 1}{z^6 + 1}$$

Options :

1. 0

2. $\frac{5}{3}$

3. 1

4. $-\frac{5}{3}$

Question Number : 4 Question Id : 464662504 Question Type : MCQ Option Shuffling : No

The Poisson distribution is often used to model

Options :

1. The number of successes in a fixed number of Bernoulli trials
2. The number of events occurring in a fixed interval of time and space
3. The distribution of continuous random variables
4. The probability of normal deviations from the mean

Question Number : 5 Question Id : 464662505 Question Type : MCQ Option Shuffling : No

If one represents the central force as $\vec{F} = F(r)\hat{r}$, then the force is attractive if

Options :

1. $F(r) < 0$
2. $F(r) > 0$
3. $F(r) = 0$
4. None of these

Question Number : 6 Question Id : 464662506 Question Type : MCQ Option Shuffling : No

In exoergic collision between two particles, the final kinetic energy is

Options :

1. exceeds the initial kinetic energy
2. equals the initial kinetic energy
3. less than the initial kinetic energy
4. bears no relation with initial kinetic energy

Choose the right form for Hamilton's equations of motion are

Options :

1.
$$\frac{\partial H}{\partial p} = \dot{q}, \quad \frac{\partial H}{\partial q} = \dot{p}$$

2.
$$\frac{\partial H}{\partial p} = -\dot{q}, \quad \frac{\partial H}{\partial q} = \dot{p}$$

3.
$$\frac{\partial H}{\partial p} = -q, \quad \frac{\partial H}{\partial q} = p$$

4.
$$\frac{\partial H}{\partial p} = \dot{q}, \quad \frac{\partial H}{\partial q} = -\dot{p}$$

The rotating earth has settled down in the form of an oblate-ellipsoid flattened at the poles due to

Options :

1. gravitational acceleration
2. coriolis acceleration
3. effective gravitational acceleration
4. centrifugal acceleration

Match the pairs between the quantities and their units.

- | | |
|-------------------|-------------------------------|
| A. Force | i. Newtons-meters per coulomb |
| B. Charge | ii. Newtons per coulomb |
| C. Electric field | iii. Coulomb |
| D. Potential | iv. Newtons |

Options :

1. A – iv B – iii C – ii D – i
2. A – i B – ii C – iii D – iv
3. A – ii B – iii C – iv D – i
4. A – i B – iii C – iv D – ii

A charged particle starts at a origin moves with a constant angular velocity in a uniform magnetic field, then which of the following statements are true ?

- a. Uniform magnetic field has no effect on the K.E. of the particle
- b. The particle gyrates in an orbit about magnetic field line
- c. This idea can be used for magnetic focusing of electron beams
- d. It is a principle used in cyclotron

Options :

- 1. Only a
- 2. Only a, b, d
- 3. Only b, c, d
- 4. All a, b, c, d

Say true or false statements.

- A. Magnetic forces may alter the direction in which a particle moves, but they cannot speed it up or slow it down.
- B. Current in a wire is the charge per unit time, passing a given point.
- C. Current is measured in coulombs.

Options :

- 1. Only statement C is false
- 2. Only statement A is false
- 3. Only statement B is false
- 4. All statements are true

A magnetic needle is kept in a non-uniform magnetic field. It experiences

Options :

1. a force and torque
2. a force but not torque
3. a torque but not force
4. neither a force nor a torque

When coupling two angular momenta $j_1 = \frac{1}{2}$ and $j_2 = \frac{1}{2}$, the possible values of total angular momenta are

Options :

1. $1, 1$

2. $0, 0$

3. $0, 1$

4. $\frac{1}{2}, \frac{1}{2}$

The transition probability per unit time from an initial state to final state due to the weak time dependent perturbation is proportional to

Options :

1. $\langle f|H|i\rangle$

2. $\rho(E_f)$

3. $|\langle f|H|i\rangle|^2 \rho(E_f)$

4. $\frac{1}{\rho(E_f)}$

In the Stern-Gerlach experiment, when a beam of silver atoms passes through an inhomogeneous magnetic field, what happens ?

Options :

1. The beam remains undivided
2. The beam splits into three components corresponding to the three possible spin-state
3. The beam splits into two distinct components corresponding to spin-up and spin-down
4. The beam splits into discrete components corresponding to the quantized spin states of electrons

Question Number : 16 Question Id : 464662516 Question Type : MCQ Option Shuffling : No

The time dependent Schrödinger equation $H\psi = E\psi$ describes the motion of

Options :

1. Relativistic material particle
2. Non-relativistic material particle
3. Relativistic as well as non-relativistic material particle
4. None of the above

Question Number : 17 Question Id : 464662517 Question Type : MCQ Option Shuffling : No

As the constraints on the system increase, the number of microstates accessible to the system will

Options :

1. Increase
2. Remains same
3. Decrease
4. Depends on types of constraints

Question Number : 18 Question Id : 464662518 Question Type : MCQ Option Shuffling : No

For a system of N particles, for a thermodynamic transformation change in N, the first law takes the form (μ = chemical potential)

Options :

1. $du = TdS + PdV + \mu dN$
2. $du = TdS + \mu dN$
3. $du = TdS - PdV + \mu dN$
4. $du = -TdS + PdV + \mu dN$

Question Number : 19 Question Id : 464662519 Question Type : MCQ Option Shuffling : No

According to which statistics there is restriction on the number of particles in a given state ?

Options :

1. Fermi-Dirac only
2. Bose-Einstein only
3. Maxwell-Boltzmann only
4. Both Fermi-Dirac and Maxwell-Boltzmann

Question Number : 20 Question Id : 464662520 Question Type : MCQ Option Shuffling : No

Grand canonical ensemble corresponds to an ensemble in which exchange of _____ is allowed.

Options :

1. Energy only
2. Particles only
3. Both energy and particles
4. None of these

From the relation $\ln Q(Z, V, T) = \frac{PV}{K_B T}$, where Q denotes the grand partition function, the expansion of P in terms of density $n (= N/V)$ and thermal wavelength λ , $\frac{PV}{NK_B T} = 1 + a_2 \lambda^3 n + a_3 (\lambda^3 n)^2 + \dots$ is termed as

Options :

1. Cluster expansion
2. Critical expansion
3. Parametric expansion
4. Virial expansion

Which of the following properties are required for a dye to ensure optimal performance and long-term stability in a Dye-Sensitized Solar Cell (DSSC) ?

- a. The dye should have luminescence properties
- b. The absorption spectra of dye must span over UV-VIS and Near-Infrared (NIR) Regions
- c. The dye periphery should be hydrophobic
- d. The highest Occupied Molecular Orbit (Homo) of the dye should lie lower than the potential of the redox electrolytes

Options :

- 1. Only a and b
- 2. Only b, c and d
- 3. Only a, b and d
- 4. All of the above

Which of the following are advantages of Quantum Dot Light Emitting Diodes ?

- a. Broad emission width
- b. High quantum yield
- c. Tunable emission wavelength
- d. High stability

Options :

- 1. Only a, b and d
- 2. Only b, c and d
- 3. Only a and c
- 4. Only b and d

For proper working of a bipolar junction transistor the _____ is always connected in reverse bias, which produces a small leakage current due to minority charge carriers.

Options :

1. Emitter base junction
2. Collector base junction
3. Emitter collector junction
4. None of the above

Question Number : 25 Question Id : 464662525 Question Type : MCQ Option Shuffling : No

A pure semiconductor without any doping is called as

Options :

1. Intrinsic semiconductor
2. Extrinsic semiconductor
3. P-type semiconductor
4. N-type semiconductor

Question Number : 26 Question Id : 464662526 Question Type : MCQ Option Shuffling : No

In quantum mechanics, the generators of SU(2) are represented by _____.

Options :

1. Pauli matrices
2. Identity matrix
3. Rotation matrices in 3D
4. Dirac Gamma matrices

For fitting data to a power law model of the form $y = Cx^k$ the least squares method is typically applied by

Options :

1. Minimizing residuals in the original non-linear form
2. Using cubic spline interpolation
3. Transforming to a linear relation $\ln y = \ln C + k \ln x$
4. Applying Fourier series expansion

In general relativity, the metric tensor $g_{\mu\nu}$ is used to :

Options :

1. Define curvature scalar directly
2. Relate covariant and contravariant components of vectors
3. Measure only time intervals
4. Replace the Stress-Energy tensor

Let a_{ij} be the elements of 3×3 matrix and A^{ij} be the corresponding cofactors. If Δ is determinant of matrix, then the relation between a_{ij} and A^{kj} is

Options :

1. $a_{ij} A^{kj} = 0$

2. $a_{ij} A^{kj} = \delta_{ij}$

3. $a_{ij} A^{kj} = \Delta \delta_i^k$

4. $a_{ij} A^{kj} = \Delta^2 \delta_i^k$

Question Number : 30 Question Id : 464662530 Question Type : MCQ Option Shuffling : No

In numerical analysis, which method is commonly used for polynomial interpolation of a given dataset ?

Options :

1. Newton's divided difference method
2. Euler's method
3. Runge-Kutta method
4. Gauss-Siedel method

Question Number : 31 Question Id : 464662531 Question Type : MCQ Option Shuffling : No

In solving partial differential equations using the finite difference method, the stability of the numerical scheme is often analyzed using :

Options :

1. Fourier Transform methods (von Neumann stability analysis)
2. Taylor series expansion
3. Gaussian Elimination
4. Laplace Transform

In electrostatics, the Green's function for Poisson equation is used to :

Options :

1. Find the eigenvalues of the Laplacian operator
2. Solve the homogeneous Laplace equation
3. Construct the potential at a point due to a given charge distribution
4. Normalize the wave function in quantum mechanics

Which of the following is a key property of the Bisection method compared to Newton-Raphson method for solving roots of equations ?

Options :

1. Faster convergence near the root
2. Guaranteed convergence if the function changes sign over the interval
3. Does not require bracketing of the root
4. Requires evaluation of derivatives

Question Number : 34 Question Id : 464662534 Question Type : MCQ Option Shuffling : No

The multiplicative constant λ is associated with which type of transformations of canonical co-ordinates ?

Options :

1. Point transformation
2. Canonical transformation
3. Scale transformation
4. Identity transformation

Question Number : 35 Question Id : 464662535 Question Type : MCQ Option Shuffling : No

In phase space, the motion of system is described by _____.

Options :

1. A scalar
2. A fixed point
3. A curve (trajectory)
4. A surface

If X and Y are two dynamical variables then Poisson bracket is defined as

Options :

1.
$$[X, Y]_{q,p} = \sum_j \left(\frac{\partial X}{\partial q_j} \frac{\partial Y}{\partial p_j} - \frac{\partial X}{\partial p_j} \frac{\partial Y}{\partial q_j} \right)$$

2.
$$[X, Y]_{q,p} = \sum_j \left(\frac{\partial X}{\partial q_j} \frac{\partial Y}{\partial p_j} + \frac{\partial X}{\partial p_j} \frac{\partial Y}{\partial q_j} \right)$$

3.
$$[X, Y]_{q,p} = \sum_j \left(\frac{\partial X}{\partial p_j} \frac{\partial Y}{\partial q_j} - \frac{\partial X}{\partial q_j} \frac{\partial Y}{\partial p_j} \right)$$

4.
$$[X, Y]_{q,p} = \sum_j \left(\frac{\partial X}{\partial q_j} \frac{\partial Y}{\partial q_j} - \frac{\partial X}{\partial p_j} \frac{\partial Y}{\partial p_j} \right)$$

Hamiltonian principal function is thus the generator of a canonical transformation to constant co-ordinates and moments. When solving Hamiltonian-Jacobi equation we are at the _____ time obtaining a solution to the mechanical problem.

Options :

1. Same
2. Different
3. Un-equal
4. None of the above

When a constant _____ is applied to the pendulum at rest, there will be a initial transient behaviour that eventually settles down to a dynamic steady state after the transients die out.

Options :

1. Torque
2. Rotational force
3. Twisting force
4. All of the above

Which are most general form of Noether's theorem of transformation of co-ordinates and time ?

Options :

1.
$$\frac{\partial L}{\partial \dot{q}_i}(\dot{q}_i X - Q_i) - LX + F = \text{constant}$$

2.
$$\frac{\partial L}{\partial \dot{q}_i}(\dot{q}_i X - Q_i) - LX + F = 0$$

3.
$$\frac{\partial L}{\partial \dot{q}_i}(\dot{q}_i X - Q_i) + LX + F = \text{constant}$$

4.
$$\frac{\partial L}{\partial \dot{q}_i}(\dot{q}_i X - Q_i) + LX + F = 0$$

Question Number : 40 Question Id : 464662540 Question Type : MCQ Option Shuffling : No

The method of separation of variables in Hamilton-Jacobi theory is applicable when _____.

Options :

1. The system is non-conservative
2. The Hamiltonian allows separable co-ordinates
3. Time is not defined
4. Momentum is zero

Question Number : 41 Question Id : 464662541 Question Type : MCQ Option Shuffling : No

In more advanced dynamical systems, stability is analyzed using

Options :

1. Generalized co-ordinates and conjugate momenta
2. Generalized co-ordinate and cyclic co-ordinate
3. Cyclic co-ordinate and conjugate momenta
4. Cyclic co-ordinate and conjugate co-ordinate

Consider a rectangular waveguide with internal dimensions 'a' (width) and 'b' (height) where $a > b$. If the waveguide operating in its dominant mode. Which of the following statements are correct ?

- a. The dominant mode is designated as TE_{10} and its cut off wavelength $\lambda_c = 2a$.
- b. The phase velocity V_p of the wave inside the guide is always zero.
- c. At the cut off frequency, the group velocity V_g , becomes zero and the wave does not transport energy.
- d. The waveguide can support TM_{10} and TM_{01} modes as long as the frequency is high enough.

Options :

- 1. Only a and b
- 2. Only a and c
- 3. Only a and d
- 4. Only a, b and c

The divergence of which tensor gives the inhomogeneous Maxwell's equations ?

- a. The Maxwell stress tensor
- b. The electromagnetic field tensor
- c. The metric tensor
- d. The dual tensor

Options :

- 1. Only a and d
- 2. Only a
- 3. Only b
- 4. Only a and c

Question Number : 44 Question Id : 464662544 Question Type : MCQ Option Shuffling : No

Electric field due to a potential varying as $V = 5x$ (V in volts and x in meters) along x -axis is

Options :

1. 5 V/m
2. -5 V/m
3. 0
4. 10 V/m

Question Number : 45 Question Id : 464662545 Question Type : MCQ Option Shuffling : No

Magnetic field inside a perfect conductor placed in an external magnetic field is

Options :

1. constant and zero
2. constant and non-zero
3. infinite
4. maximum at surface and minimum at center

Cerenkov radiation is observed as a faint bluish light when the speed of a penetrating particle exceeds which of the following ?

- a. The speed of light in that medium
- b. The phase velocity of penetrating particle in that medium
- c. The group velocity of light in that medium
- d. The speed of light in vacuum

Options :

- 1. Only a and b
- 2. Only c
- 3. Only d
- 4. Only b and d

Question Number : 47 Question Id : 464662547 Question Type : MCQ Option Shuffling : No

In a wire carrying steady current, the term $\mathbf{J} \cdot \mathbf{E}$ in Poynting's theorem is

Options :

1. Zero
2. Negative
3. Positive
4. Purely imaginary

Write true or false statements.

- A. The energy flux density of wave is energy per unit area, per unit time.
- B. This energy (S) transported by the field is given by $S = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$.
- C. Electromagnetic fields carry energy and momentum.

Options :

- 1. All statements are false
- 2. All statements are true
- 3. Statements A and B are true
- 4. Only statement A is true

Say true or false statements.

- A. When we shake a string, then the waves travels down are called transverse waves.
- B. In transverse waves, displacement of the particle is perpendicular to direction of propagation.
- C. In longitudinal waves, the displacement from equilibrium is perpendicular to the direction of propagation.

Options :

- 1. Only statement C is false
- 2. Only statement A is false
- 3. Only statement B is false
- 4. Statements A and B are false

In quantum scattering theory, the differential cross section $\frac{d\sigma}{d\Omega}$ is related to the scattering amplitudes $f(\theta, \phi)$ by which of the following expressions ?

Options :

1. $\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|$

2. $\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2$

3. $\frac{d\sigma}{d\Omega} = \left| \frac{df}{d\theta} \right|^2$

4. $\frac{d\sigma}{d\Omega} = 4\pi |f(\theta, \phi)|^2$

The relativistic energy-momentum relation is given by _____.

Options :

1. $E = pc$

2. $E = \frac{p^2}{2m}$

3. $E = mc^2 + \frac{p^2}{2m}$

4. $E^2 = p^2c^2 + m^2c^4$

Question Number : 52 Question Id : 464662552 Question Type : MCQ Option Shuffling : No

In partial wave analysis, a plane wave is equivalent to

Options :

1. One spherical wave
2. Two spherical waves
3. A number of spherical waves
4. Standing waves only

Assertion (A) : The splitting of energy states having the same principal quantum number 'n' is called fine structure.

Reason (R) : Energy levels having the same 'n' split due to spin-orbit interaction and relativistic corrections.

Options :

1. Both (A) and (R) are true and (R) is correct explanation
2. Both (A) and (R) are true, but (R) is not correct explanation
3. (A) is true, but (R) is false
4. (A) is false, but (R) is true

The scattering amplitude in born approximation is proportional to _____.

Options :

1. Laplacian of potential
2. Fourier transform of potential
3. Gradient of potential
4. Both Laplacian and gradient of potential

The Dirac equation may be written in the precisely same form as non-relativistic schrodinger equation as

Options :

$$\hat{H}\psi = -i\hbar \frac{\partial \psi}{\partial t}$$

1.

$$\hat{H}\psi = i\hbar \frac{\partial \psi}{\partial t}$$

2.

$$\hat{H}\psi = \pm i\hbar \frac{\partial \psi}{\partial t}$$

3.

all of the above

4.

If the scattering amplitude is purely real, the optical theorem will imply _____.

Options :

1. the total scattering cross section is zero
2. the total scattering cross section is maximum
3. the phase shift is 90°
4. none of the above

The change in momentum of the particle over a wavelength must be much less than the momentum itself in order for WKB approximation to be valid. Since P changes rapidly in regions where $\left| \frac{dv}{dx} \right|$ is _____, the WKB approximation cannot be expected to hold at points where $v(x)$ varies rapidly.

Options :

1. large
2. small
3. middle
4. none of the above

Which of the following is a real-world manifestation of Bose-Einstein condensation ?

Options :

1. Ferromagnetism in iron
2. Superfluidity in liquid helium (He^4)
3. Superconductivity in metals
4. Classical ideal gas behaviour

Which of the following statements correctly distinguishes diamagnetism from paramagnetism ?

Options :

1. Diamagnetism arises from unpaired electron spins while paramagnetism arises from paired spins
2. Diamagnetism is temperature dependent while paramagnetism is not
3. Diamagnetism originates from induced currents opposing the applied field while paramagnetism arises from permanent magnetic moments aligning with the applied field
4. Both diamagnetism and paramagnetism vanish at high temperatures

In the 1D-Ising model without an external field, what is the nature of the phase transition at finite temperature ?

Options :

1. First order transition with latent heat
2. Second order transition at critical temperature
3. No phase transition occurs at finite temperature
4. Continuous transition with spontaneous magnetization

For a particle performing successive displacements in one dimension of length l , after a total of N such steps, the particle is located at $x = ml$, where $-N \leq m \leq N$ (m : integer). If n_1 and n_2 denote the number of steps to the right and left respectively, $N = n_1 + n_2$; the different possible ways of taking the n_1 and n_2 steps are :

Options :

1. $N!$

2. $\frac{n_1 + n_2}{N}$

3. $n_1!n_2!$

4. $\frac{N!}{n_1!n_2!}$

The temperature at which ferromagnetic material becomes paramagnetic material is _____ temperature.

Options :

1. Langevin
2. Brillouin
3. Curie
4. Weiss

Question Number : 63 Question Id : 464662563 Question Type : MCQ Option Shuffling : No

The Curie law of paramagnetism holds good for _____.

Options :

1. $\mu B = KT$
2. $\mu B \gg KT$
3. $\mu B \ll KT$
4. $\mu B = \frac{11}{KT}$

Question Number : 64 Question Id : 464662564 Question Type : MCQ Option Shuffling : No

The example/s of second order phase transitions are

Options :

1. Transition of liquid Helium I to liquid Helium II, at λ point (2.19 K)
2. Transition of ferromagnetic material to paramagnetic material at Curie point
3. Both (1) and (2)
4. None of the above

Question Number : 65 Question Id : 464662565 Question Type : MCQ Option Shuffling : No

The phrase “condensation in momentum space” is used for Bose-Einstein condensation mainly to emphasize that

Options :

1. The condensation occurs only in real space
2. The origin lies in the symmetry of the wave function
3. It is different from gas-liquid transition
4. The particles interact strongly at low temperatures

Question Number : 66 Question Id : 464662566 Question Type : MCQ Option Shuffling : No

A lock-in detector is used to

Options :

1. Generate high frequency signals
2. Detect weak signals buried in noise
3. Amplify power only
4. Store electrical energy

Which two of the following are consequences of applying a nonlinear regression to smoothed data ?

- a. The computed sum-of-squares will over estimate the true scatter
- b. Residuals will no longer be independent of one another !
- c. Standard errors for the parameters will be calculated as too small
- d. Model comparisons will become highly accurate and valid

Options :

- 1. Only a and c
- 2. Only b and c
- 3. Only b and d
- 4. Only a and d

Question Number : 68 Question Id : 464662568 Question Type : MCQ Option Shuffling : No

The linear regression method aims to minimize

Options :

1. The absolute difference between average values
2. The sum of square of errors
3. The sum of errors
4. None of the above

Which of the following statements accurately describe the trade-offs between the resistive and capacitive, compensated and uncompensated instrumentation amplifier architecture ?

- a. The capacitive uncompensated architecture achieves maximum DC headroom but suffers from poor noise performance and lower CMRR.
- b. Applying compensation to a resistive architecture improves CMRR and noise performance while preserving high input impedance.
- c. The capacitive compensated architecture offers both maximum DC headroom and high input impedance.
- d. Both resistive and capacitive uncompensated configuration exhibit superior noise performance compared to their compensated counterparts.

Options :

- 1. All of the above
- 2. Only a
- 3. Only d
- 4. Only b and c

The transducers that converts the input signal into the output signal, which is a discrete function of time is known as _____ transducers.

Options :

1. Active
2. Analog
3. Digital
4. Pulse

In an Op-Amp based two stage instrumentation amplifier which of the following is true for first stage ?

- a. The first stage, amplifies the differential signal at the input
- b. The first stage, restricts the common mode to unity
- c. The first stage, comprises two amplifiers
- d. The first stage has higher input impedance

Options :

- 1. Only a, b
- 2. Only a, b, c
- 3. Only b, c, d
- 4. Only a, c, d

In case of differential amplifier, the ratio of differential voltage gain to common mode voltage gain is called

Options :

1. Input impedance
2. Bandwidth
3. Common mode rejection ratio
4. Slew rate

In case of first order low pass filter, the cutoff frequency f_c is given by

Options :

1. $\frac{1}{2\pi\sqrt{RC}}$

2. $\frac{1}{2\pi RC}$

3. $\frac{2\pi}{RC}$

4. $\frac{2\pi}{\sqrt{RC}}$

Question Number : 74 Question Id : 464662574 Question Type : MCQ Option Shuffling : No

The spectrum of excited helium atom is expected to be similar to that of

Options :

1. Hydrogen
2. Helium
3. Lithium
4. Sodium

Question Number : 75 Question Id : 464662575 Question Type : MCQ Option Shuffling : No

In an optical pumping system, atoms absorb photons of the frequency 5×10^{14} Hz. The energy absorbed by one atom is (Given : $h = 6.63 \times 10^{-34}$ JS)

Options :

1. 6.6×10^{-20} J
2. 3.3×10^{-19} J
3. 5.0×10^{-21} J
4. 1.0×10^{-18} J

Question Number : 76 Question Id : 464662576 Question Type : MCQ Option Shuffling : No

The Born-Oppenheimer approximation is used to separate

Options :

1. Atomic and molecular energy levels
2. Electronic and nuclear wave functions
3. Translational and rotational motion
4. Rotational and vibrational motion

Question Number : 77 Question Id : 464662577 Question Type : MCQ Option Shuffling : No

Magnitude of electron spin angular momentum is

Options :

1. $\hbar/2$
2. 0
3. $\frac{\sqrt{3}}{2}\hbar$
4. $\sqrt{2}\hbar$

Question Number : 78 Question Id : 464662578 Question Type : MCQ Option Shuffling : No

Russell-Saunders's coupling is also termed as _____.

Options :

1. JJ coupling
2. LS coupling
3. JS coupling
4. LJ coupling

In Physics, Atomic and Molecular Physics :

Say 'True', or 'False', statements.

- A. In "Stark effect splitting of spectral lines takes place" in presence of electric field.
- B. This splitting are not symmetrical about the original line.
- C. This splitting increases linearly with field strength, known as linear Stark effect.

Options :

- 1. All statements are true
- 2. All statements are false
- 3. Only statement C is true
- 4. Only statement A is true

Population inversion is a state where

Options :

1. there are same number of atoms in excited state and lower state
2. there are less number of atoms in excited state than lower state
3. there are more number of atoms in excited state than lower state
4. number of atoms in lower state is zero

Question Number : 81 Question Id : 464662581 Question Type : MCQ Option Shuffling : No

An excitation line of wavelength $5000 \text{ \AA}$ produces a Raman shift of 4000 cm^{-1} .
Find the wavelength of the stokes line.

Options :

1. $6250 \text{ \AA}$

2. $5000 \text{ \AA}$

3. $4000 \text{ \AA}$

4. $7000 \text{ \AA}$

In Physics, Atomic and Molecular Physics : –

Match the pairs.

A. Attenuation

i. Make light rays parallel

B. Collimate

ii. Reduction of light intensity or loss

C. Beam divergence

iii. Beam spreads with increase in length

Options :

1. A – ii B – i C – iii

2. A – i B – ii C – iii

3. A – iii B – ii C – i

4. A – ii B – iii C – i

Question Number : 83 Question Id : 464662583 Question Type : MCQ Option Shuffling : No

Edge dislocation is an example of _____.

Options :

1. Line defect
2. Point defect
3. Surface defect
4. Volume defect

Which of the following statements are correct regarding energy bands and primitive cells in solids ?

- a. A single primitive cell contributes exactly two independent values of 'k' to a given energy band due to the two independent spin orientations.
- b. An energy band in crystal containing N primitive cells possesses a total capacity of 2N independent orbitals.
- c. If every primitive cell in a crystal contains exactly one atom with a valence of one, the corresponding energy band will be half filled.
- d. A band can exactly filled only if each individual atom in the cell contributes precisely two valence electrons.

Options :

- 1. Only a
- 2. Only b and c
- 3. Only a and c
- 4. Only a and d

Which of the following accurately describe physical characteristics and structural requirements of a substance in the liquid crystal state ?

- a. It is an intermediate state combining fluid behaviour with anisotropic properties.
- b. Formation require directional, rod or disc-shaped molecular geometries.
- c. It transitions through intermediate structural mesophases before melting into an isotropic liquid.
- d. Phase transitions are driven by temperature changes or solvent chemical interactions.

Options :

- 1. Only a and c
- 2. Only b, c and d
- 3. Only a, b and c
- 4. All of the above

Write the True or False statements.

- A. In ionic bond the force is due to electrostatic attractions of oppositely charged ions.
- B. In covalent bond the force is due to mutually sharing of valence electrons between atoms.
- C. Metallic bonds are due to attraction between the ion cores and conduction electron gas.

Options :

- 1. All statements are true
- 2. Only statement 'B' is true
- 3. Only statement 'C' is true
- 4. Only statement 'A' is true

Question Number : 87 Question Id : 464662587 Question Type : MCQ Option Shuffling : No

Calculate the interplanar distance (d) when X-rays are diffracted at 30° angle having $1.2304 \text{ \AA}$ wavelength for the first order diffraction.

Options :

1. 1.2304 mm.
2. 1.2304 cm.
3. 1.2304 nm.
4. $1.2304 \text{ \AA}$

Choose correct statements.

- A. Metallic conductors obey Ohm's law
- B. Metals have high electrical conductivity
- C. Metals have low thermal conductivity
- D. The conductivity of metals varies in presence of magnetic field

Options :

1. All statements are correct
2. Only statements A, B and D are correct
3. Only statement C is correct
4. Only statement D is correct

Say True or False statements.

- A. The energy of a lattice vibrations is quantised.
- B. The quantum of energy of lattice vibrations is called “phonon”.
- C. Thermally vibrations in crystals are thermally excited photons.

Options :

- 1. Only statement C is false
- 2. Statement A is false
- 3. Statement B is false
- 4. Both statements A and B are false

Question Number : 90 Question Id : 464662590 Question Type : MCQ Option Shuffling : No

For X-rays of wavelength (λ), $1\text{\AA}$ incident on a crystal with interplanar spacing (d), $2\text{\AA}$. The Bragg angle for second-order diffraction is

Options :

1. 14.5°
2. 30°
3. 45°
4. 60°

Question Number : 91 Question Id : 464662591 Question Type : MCQ Option Shuffling : No

The ratio of thermal conductivity to electrical conductivity in metals is proportional to absolute temperature

Options :

1. Wiedemann-Franz law
2. Ohm's law
3. Stefan's law
4. Bragg's law

If quarks are described as “the smallest elementary building blocks of matter with fractional electric charge”, which of the following best evaluate the implications of this definition for our understanding of atomic structure ?

Options :

1. It implies that atoms are the ultimate indivisible units of matter, as in Dalton's theory
2. It redefines the term “ elementary particle” suggesting that even protons and neutrons have internal structure made of quarks
3. It shows that quarks combine to form electrons and positrons
4. It suggests that quarks can exist freely outside atomic nuclei

The binding energy per nuclear (B/A) against mass number graph gives following results.

- a. The value of binding energy per nucleon vary in an erratic manner for very light nuclides.
- b. For $A > 20$ the binding energy per nucleon is remarkably constant.
- c. The curve indicates that B/A increases to a maximum of about 9.8 MeV per nucleon.
- d. Discontinuities were observed equal to magic numbers in the curve.

Which of the statement given above is/are correct ?

Options :

- 1. Only a and c
- 2. Only d
- 3. a, b and c

4. a, b and d

Question Number : 94 Question Id : 464662594 Question Type : MCQ Option Shuffling : No

Which of the following option is one of the evidence of nuclear shell structure ?

Options :

1. The nuclei corresponding to nuclear magic number have less stable isotopes

2. Binding energy per nucleon curve shows continuity at nuclear magic number

3. Nuclei above a closed shell structure have a negative quadrupole moment and those below have positive moments

4. Nuclei with N or Z number equal to nuclear magic number are comparatively unstable

According to C, P and T invariance

Options :

1. The strong and electromagnetic interactions are invariant under the separate operations of C, P and T
2. The weak interaction conserve parity
3. The weak interaction is invariant under charge conjugation
4. Under time reversal operation displacement, acceleration and electric fields are not invariant

Question Number : 96 Question Id : 464662596 Question Type : MCQ Option Shuffling : No

A reaction in which the projectile becomes loosely bound in the nucleus and shares its energy with all nuclear constituents is called

Options :

1. Direct reaction
2. Elastic scattering
3. Compound nucleus reaction
4. Beta decay

Question Number : 97 Question Id : 464662597 Question Type : MCQ Option Shuffling : No

Which interactions are invariant under parity ?

Options :

1. Weak interaction only
2. Strong and electromagnetic interaction
3. Strong and weak interactions
4. All fundamental interactions

Which of the following is true regarding a halogen quenched G.M. Tube ?

Options :

1. It works only for gamma rays
2. It gives nearly infinite operation life
3. It reduces sensitivity
4. It increases dead time

In semi-empirical mass formula, match the types of nuclear energy with its dependence.

List – 1

- A. Volume energy
- B. Surface energy
- C. Coloumb energy
- D. Asymmetry energy

List – 2

- 1. A
- 2. $A^{2/3}$
- 3. $Z^2/A^{1/3}$
- 4. $\frac{(N-Z)^2}{A}$

Options :

1. A – 1 B – 2 C – 3 D – 4

2. A – 2 B – 1 C – 3 D – 4

3. A – 1 B – 3 C – 2 D – 4

4. A – 1 B – 2 C – 4 D – 3

The shape of nucleus is taken _____ because for a given volume, this shape possesses the least surface area and thus provides maximum short range binding forces between nucleons in the nucleus.

Options :

1. Spherical
2. Cylindrical
3. Elliptical
4. Plane